

Markscheme

Mathematics: analysis and approaches
Practice question bank

109. (a)

In the expansion of $(1 + x)^n$, the coefficients of x^2 and x^3 are in the ratio $1 : 3$.

Find the value of n .

$$\binom{n}{2} : \binom{n}{3} = 1 : 3 \Rightarrow \binom{n}{3} = 3\binom{n}{2} \quad (M1)$$

$$\frac{n(n-1)(n-2)}{6} = 3 \times \frac{n(n-1)}{2} \quad (M1) \text{ A1}$$

$$\text{simplify: } \frac{n-2}{6} = \frac{3}{2} \Rightarrow n-2 = 9 \quad (M1)$$

$$n = 11 \quad \text{A1}$$

[5 marks]